PythonRobotics: a Python code collection of robotics algorithms

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Abstract

This paper describes an Open Source Software (OSS) project: PythonRobotics[21]. This is a collection of robotics algorithms implemented in the Python programming language. The focus of the project is on autonomous navigation, and the goal is for beginners in robotics to understand the basic ideas behind each algorithm. In this project, the algorithms which are practical and widely used in both academia and industry are selected. Each sample code is written in Python3 and only depends on some standard modules for readability and ease of use. It includes intuitive animations to understand the behavior of the simulation.

1 Introduction

In recent years, autonomous navigation technologies have received huge attention in many fields. Such fields include, autonomous driving[22], drone flight navigation, and other transportation systems.

An autonomous navigation system is a system that can move to a goal over long periods of time without any external control by an operator. The system requires a wide range of technologies: It needs to know where it is (localization), where it is safe (mapping), where and how to move (path planning), and how to control its motion (path following). The autonomous system would not work correctly if any of these technologies is missing.

Educational materials are becoming more and more important for future developers to learn basic autonomous navigation technologies. Because these autonomous technologies need different technological skill sets such as: linear algebra, statistics, probability theory, optimization theory, and control theory etc. It needs a lot of time to be familiar with these

PythonRobotics：机器人算法的 Python 代码集合

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抽象的

本文描述了一个开源软件（OSS）项目：PythonRobotics[21]。这是用 Python 编程语言实现的机器人算法的集合。该项目的重点是自主导航，目标是让机器人初学者了解每种算法背后的基本思想。本项目选择了学术界和工业界实用且广泛使用的算法。每个示例代码都是用Python3编写的，并且仅依赖于一些标准模块以提高可读性和易用性。它包括直观的动画来理解模拟的行为。

1 简介

近年来，自主导航技术在许多领域受到了广泛关注。这些领域包括自动驾驶[22]、无人机飞行导航和其他交通系统。

自主导航系统是一种无需操作员任何外部控制即可长时间移动到目标的系统。该系统需要广泛的技术：它需要知道它在哪里（定位）、哪里安全（绘图）、在哪里以及如何移动（路径规划）以及如何控制其运动（路径跟踪）。如果缺少其中任何一项技术，自治系统将无法正常工作。

对于未来开发人员学习基本的自主导航技术来说，教材变得越来越重要。因为这些自主技术需要不同的技术技能，例如：线性代数、统计学、概率论、最优化理论和控制论等，需要大量的时间来熟悉这些

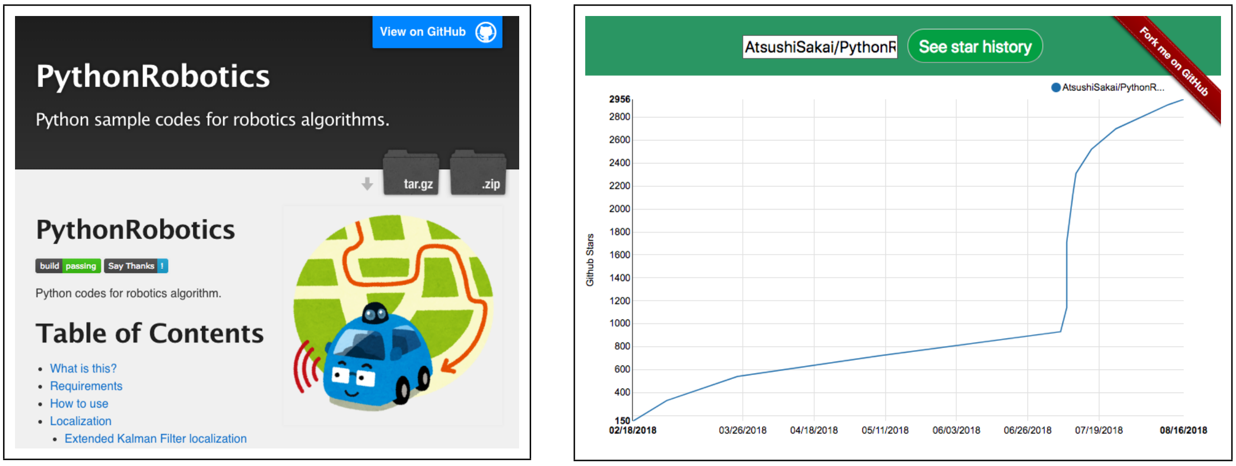


Figure 1: Left: PythonRobotics Project page, Right: GitHub star history graph using [16]

technological areas. Therefore, good educational resources for learning basic autonomous navigation technologies are needed. Our project which is described in this paper aims to be one such resource.

In this paper, an Open Source Software(OSS) project: PythonRobotics[21] is described. This project provides a code collection of robotics algorithms, especially focusing on autonomous navigation. The principle goal is to provide beginners with the tools necessary to understand it. It is written in Python[12] under MIT license[7]. It has a lot of simulation animations that shows behaviors of each algorithm. It helps for learners to understand its fundamental ideas. The readers can find the animations in the project page `https://atsushisakai.github.io/PythonRobotics/`. The left figure in Fig.1 shows the front image of the project page. All source codes of this project are provided at `https://github.com/AtsushiSakai/PythonRobotics`.

This project was started from Mar. 21 2016 as a self-learning project. It already has over 3000 stars on GitHub, and the right figure in Fig.1 shows the history graph of the star counts using [16].

This paper is organized as follows: Section 2 reviews the related works. The philosophy of this project is presented in Section 3. The repository structure and some technical backgrounds and simulation results are provided in Section 4. Conclusions are drawn and some future works are presented in Section 5. Section 6 acknowledges for contributors and supporters.

2 Related works

There are already some great educational materials for learning autonomous navigation technologies.

S. Thrun et al. wrote a great textbook "Probabilistic robotics" which is a bible of localization, mapping, and Simultaneous Localization And Mapping (SLAM) for mobile robotics[31]. E. Frazzoli et al. wrote a great survey paper about path planning and control techniques for autonomous driving [25]. G. Bautista et al. wrote a survey paper focusing on path planning for automated vehicles[22]. J. Levinson wrote an overview paper about systems and algorithms towards fully autonomous driving[24].

These papers help readers to learn state-of-the-art autonomous navigation technologies. However, it might be difficult for beginners to understand the basic ideas of the technologies and algorithms because the papers don't include implementation examples.

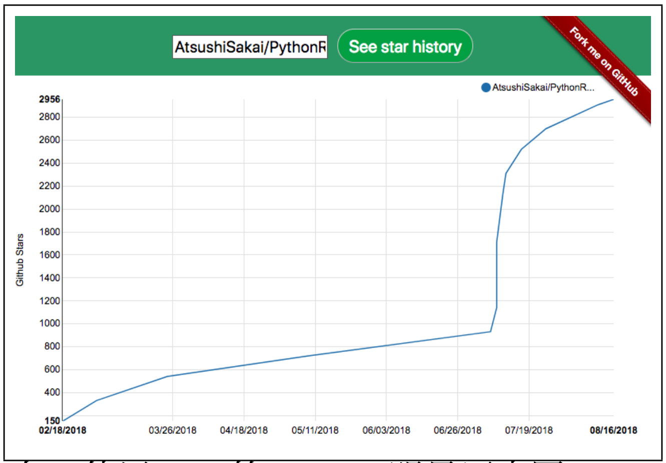


图 1：左：PythonRobotics 项目页面，右：使用 [16] 的 GitHub 明星历史图

技术领域。因此，需要良好的教育资源来学习基本的自主导航技术。本文中描述的我们的项目旨在成为这样的资源之一。

本文描述了一个开源软件（OSS）项目：PythonRobotics[21]。该项目提供了机器人算法的代码集合，特别关注自主导航。其主要目标是为初学者提供理解它所需的工具。它是在 MIT 许可 [7] 下用 Python [12] 编写的。它有很多模拟动画来显示每种算法的行为。它有助于学习者理解其基本思想。读者可以在项目页面https://atsushisakai.github.io/PythonRobotics/中找到动画。图1左图为项目页面的正面图片。该项目的所有源代码均在https://github.com/AtsushiSakai/PythonRobotics 中提供。

该项目作为自学项目于2016年3月21日启动。它在 GitHub 上已经有超过 3000 颗星，图 1 右图显示了使用[16]的星数历史图。

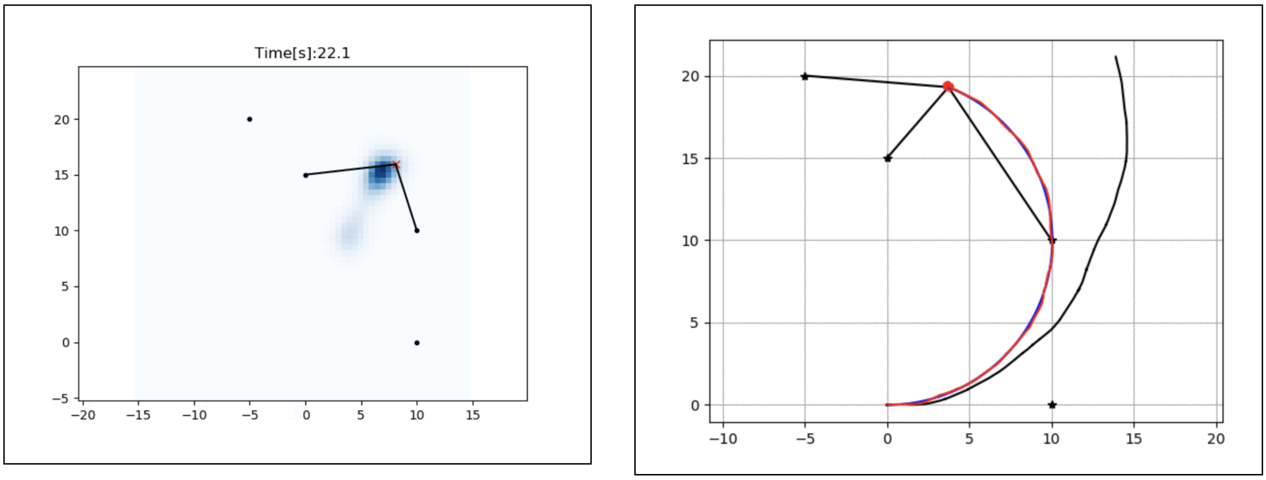
本文的结构如下：第二部分回顾了相关工作。第 3 节介绍了该项目的理念。第 4 节提供了存储库结构以及一些技术背景和模拟结果。第 5 节给出了结论并介绍了一些未来的工作。第 6 节向贡献者和支持者致谢。

2 相关作品

已经有一些很棒的学习自主导航技术的教育材料。

S.特伦等人。写了一本很棒的教科书《概率机器人学》，它是移动机器人定位、建图以及同步定位和建图（SLAM）的圣经[31]。 E.弗拉佐利等人。写了一篇关于自动驾驶路径规划和控制技术的精彩调查论文[25]。 G.包蒂斯塔等人。撰写了一篇专注于自动车辆路径规划的调查论文[22]。 J. Levinson 撰写了一篇关于完全自动驾驶的系统和算法的概述论文[24]。

这些论文帮助读者学习最先进的自主导航技术。然而，由于论文不包含实现示例，初学者可能很难理解技术和算法的基本思想。



Histogram filter localization Particle filter localization

Figure 2: Localization simulation results

Many universities provide great classes to learn robotics and share their lecture notes. For example, University of Freiburg provides an introduction class to mobile robotics[2]. Swiss Federal Institute of Technology in Zurich (ETH in Zurich) also provides a class about autonomous mobile robot[3] and robotics programming with Robot Operating System(ROS)[4].

Like the academic literature referenced, these lecture notes also help readers to learn autonomous navigation technologies. However, it might be also difficult for beginners of robotics to understand the basic ideas of the technologies and algorithms if the reader is not a student of the class, because these lecture notes doesn't include actual implementation examples of robotics algorithms.

Robot Operating System (ROS) is a middle-ware for robotics software development[13][28]. ROS initially released in 2007, and it has become the defacto standard platform in robotics software development. ROS includes basic navigation software such as the Adaptive Monte Carlo Localization (AMCL) package and the local path planner based on Dynamic Window Approach, etc[14]. Many autonomous navigation packages using ROS are also open-sourced.

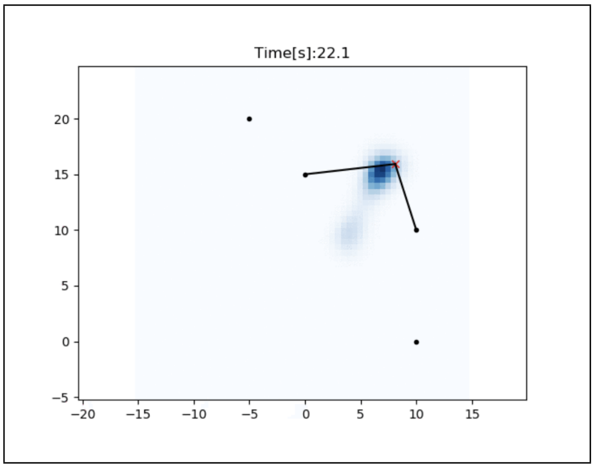
ROS packages can also be useful to learn autonomous navigation algorithms, however the documentation might be inconsistent and it can be difficult to find implementation details about the algorithms used. ROS packages are usually written in C++ because the focus is computational efficiency and not readability, which makes learning from the code harder.

Udacity Inc, which is an online learning company founded by S. Thrun, et al, is providing great online courses for autonomous navigation and autonomous driving [17]. These courses provide not only technical lectures, but also programming homework using Python. This leads to a great learning strategy for beginners to understand the technical background required for autonomous systems.

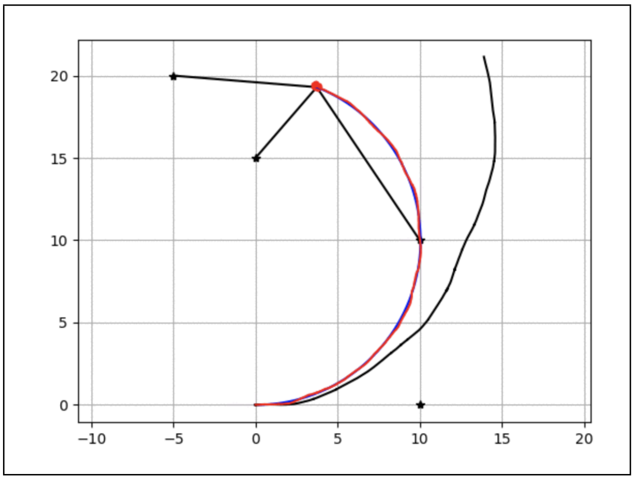
The PythonRobotics project seeks to extend Udacity's approach by giving beginners the necessary tools to understand and make further study into the methods of autonomous navigation. The details of concepts of this OSS project will be described in the next section.

3 Philosophy

In this section, the philosophy of this project is described. The PythonRobotics project is based on three main philosophies. Each philosophy will be discussed separately in this



Histogram filter localization



Particle filter localization

图2：定位仿真结果

许多大学提供精彩的课程来学习机器人技术并分享他们的讲义。例如，弗莱堡大学提供移动机器人技术入门课程[2]。苏黎世瑞士联邦理工学院（苏黎世联邦理工学院）还提供了有关自主移动机器人[3]和使用机器人操作系统（ROS）[4]进行机器人编程的课程。

与引用的学术文献一样，这些讲义也可以帮助读者学习自主导航技术。然而，如果读者不是该课程的学生，机器人初学者也可能很难理解技术和算法的基本思想，因为这些讲义不包括机器人算法的实际实现示例。

机器人操作系统（ROS）是用于机器人软件开发的中间件[13][28]。 ROS 最初于 2007 年发布，现已成为机器人软件开发领域事实上的标准平台。 ROS包括自适应蒙特卡罗定位（AMCL）包和基于动态窗口方法的本地路径规划器等基本导航软件[14]。许多使用 ROS 的自主导航包也是开源的。

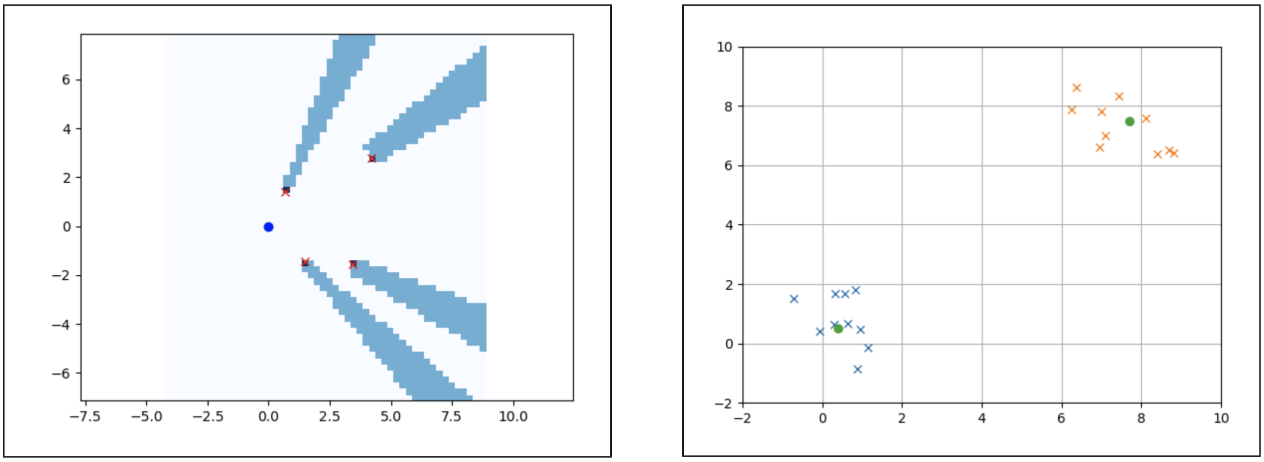
ROS 包对于学习自主导航算法也很有用，但是文档可能不一致，并且很难找到有关所用算法的实现细节。 ROS 包通常用 C++ 编写，因为重点是计算效率而不是可读性，这使得从代码中学习变得更加困难。

Udacity Inc 是一家由 S. Thrun 等人创立的在线学习公司，为自主导航和自动驾驶提供优秀的在线课程[17]。这些课程不仅提供技术讲座，还提供使用Python的编程作业。这为初学者提供了一个很好的学习策略，可以帮助他们了解自主系统所需的技术背景。

PythonRobotics 项目旨在通过为初学者提供必要的工具来理解和进一步研究自主导航方法来扩展 Udacity 的方法。该 OSS 项目的概念细节将在下一节中描述。

3 理念

在本节中，描述了该项目的理念。 PythonRobotics 项目基于三个主要理念。每个哲学将在本文中单独讨论



Grid mapping with 2D ray casting 2D object clustering with k-means algorithm

Figure 3: Mapping simulation results

section.

3.1 Readability

The first philosophy is that the code has to be easy to read. If the code is not easy to read, it would be difficult to achieve our goal of allowing beginners to understand the algorithms. Python[12] programming language is adopted in this project. Python has great libraries for matrix operation, mathematical and scientific operation, and visualization, which makes code more readable because such operations don't need to be re-implemented. Having the core Python packages allows the user to focus on the algorithms, rather than the implementations.

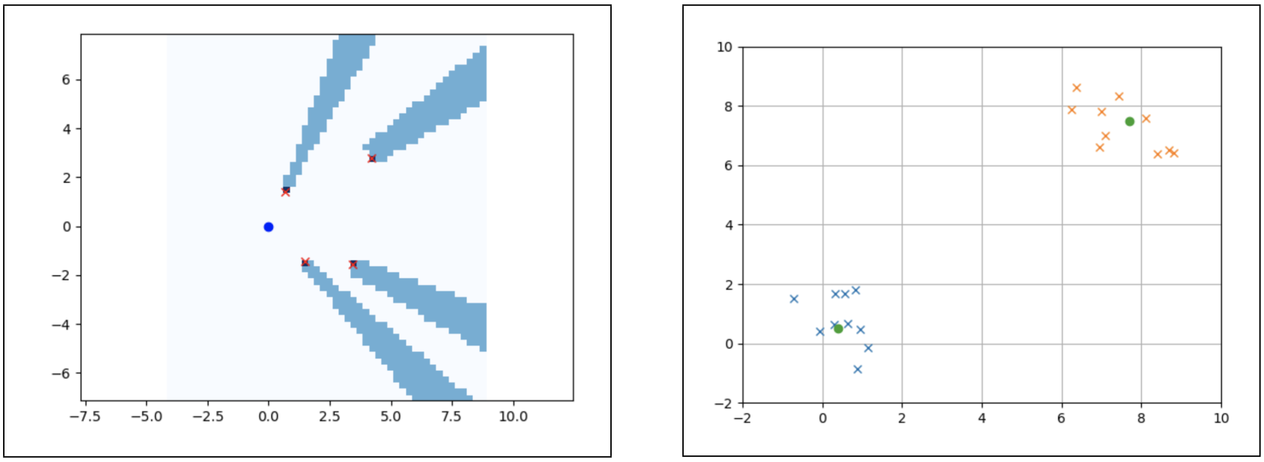
3.2 Practicality

The second philosophy is that implemented algorithms have to be practical and widely used in both academia and industry. We believe learning these algorithms will be useful in many applications. For example, Kalman filters and particle filter for localization, grid mapping for mapping, dynamic programming based approaches and sampling based approaches for path planning, and optimal control based approach for path tracking. These algorithms are implemented in this project.

3.3 Minimal dependencies

The last philosophy is minimal dependencies. Having few external dependencies allows us to run code samples easily and to convert the Python codes to other programming languages, such as C++ or Java for more practical application. Each sample code only depends some modules on Python3 as bellow.

- numpy[8] for matrix and vector operations
- scipy[15] for mathematical, scientific computing
- matplotlib[6] for plotting and visualization
- pandas[10] for data import and manipulation.



Grid mapping with 2D ray casting　　2D object clustering with k-means algorithm

图 3：映射模拟结果

部分。

3.1 可读性

第一个理念是代码必须易于阅读。如果代码不好读，就很难达到我们让初学者理解算法的目的。本项目采用Python[12]编程语言。 Python 拥有用于矩阵运算、数学和科学运算以及可视化的强大库，这使得代码更具可读性，因为此类运算不需要重新实现。拥有核心 Python 包可以让用户专注于算法，而不是实现。

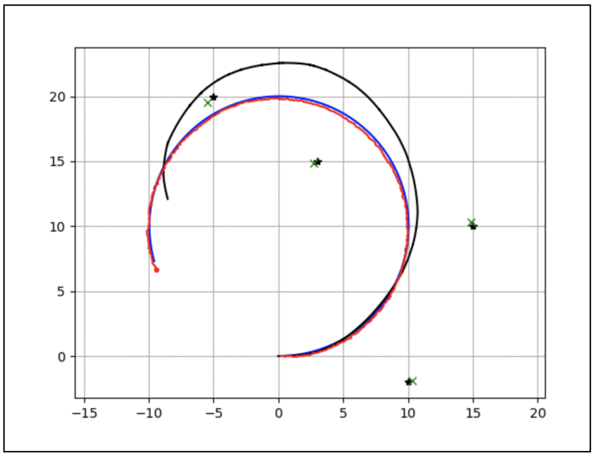
3.2 实用性

第二个理念是，实现的算法必须实用并在学术界和工业界广泛使用。我们相信学习这些算法将在许多应用中有用。例如，用于定位的卡尔曼滤波器和粒子滤波器、用于映射的网格映射、用于路径规划的基于动态规划的方法和基于采样的方法、以及用于路径跟踪的基于最优控制的方法。这些算法均在本项目中实现。

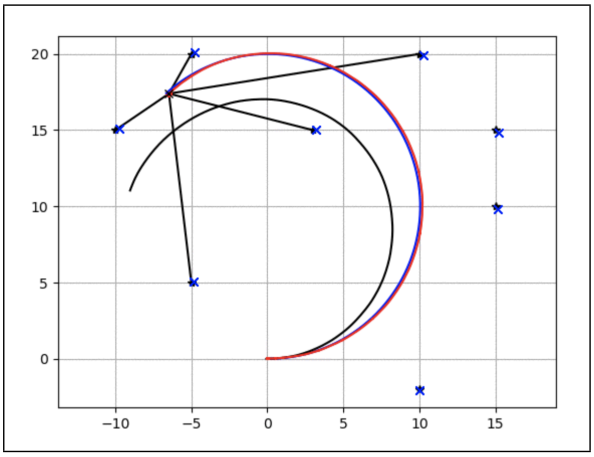
3.3 最小依赖

最后一个理念是最小依赖。由于外部依赖较少，我们可以轻松运行代码示例，并将 Python 代码转换为其他编程语言，例如 C++ 或 Java，以实现更实际的应用。每个示例代码仅依赖于Python3上的一些模块，如下所示。

- numpy[8] 用于矩阵和向量运算
- scipy[15] 用于数学、科学计算
- matplotlib[6] 用于绘图和可视化
- pandas[10] 用于数据导入和操作。



Extended Kalman filter based SLAM



FastSLAM 2.0 based SLAM

Figure 4: SLAM simulation results

- cvxpy[5] for convex optimization

These modules are OSS and can also be used for free. This repository software doesn't depend on any commercial software.

4 Repository structure

In this section, the brief repository structure is described.

There are five directories, each one corresponding to a different technical category in autonomous navigation: localization, mapping, SLAM, path planning, and path tracking. There is one sub-directory per algorithm, which has a sample code implementing and testing the algorithm. In the following subsections, some algorithms and some simulation examples in each category are described.

4.1 Localization

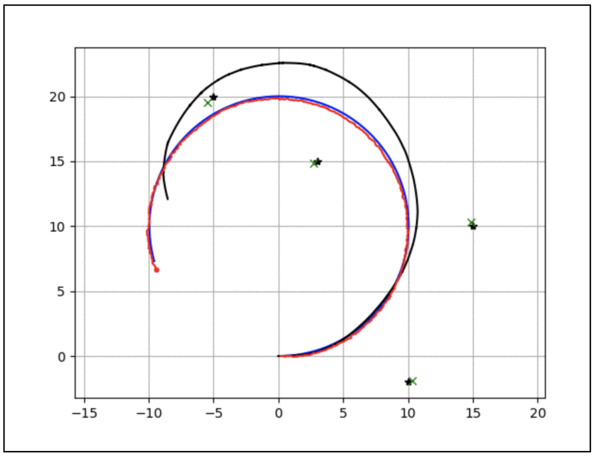
Localization is the ability of a robot to know its position and orientation with sensors such as Global Navigation Satellite System:GNSS etc. In localization, Bayesian filters such as Kalman filters, histogram filter, and particle filter are widely used[31]. Fig.2 shows localization simulations using histogram filter and particle filter.

4.2 Mapping

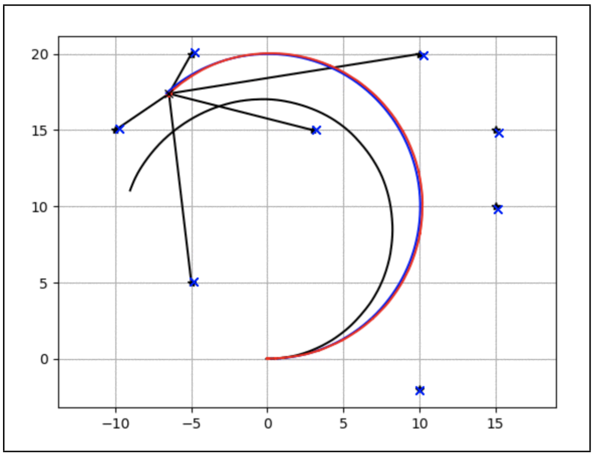
Mapping is the ability of a robot to understand its surroundings with external sensors such as LIDAR and camera. Robots have to recognize the position and shape of obstacles to avoid them. In mapping, grid mapping and machine learning algorithms are widely used[31][18]. Fig.3 shows mapping simulation results using grid mapping with 2D ray casting and 2D object clustering with k-means algorithm.

4.3 SLAM

Simultaneous Localization and Mapping (SLAM) is an ability to estimate the pose of a robot and the map of the environment at the same time. The SLAM problem is hard to



Extended Kalman filter based SLAM



FastSLAM 2.0 based SLAM

图4：SLAM仿真结果

- cvxpy[5] 用于凸优化

这些模块都是OSS，也可以免费使用。该存储库软件不依赖于任何商业软件。

4 存储库结构

在本节中，描述了简要的存储库结构。

有五个目录，每个目录对应自主导航中的不同技术类别：定位、建图、SLAM、路径规划和路径跟踪。每个算法都有一个子目录，其中包含实现和测试该算法的示例代码。在以下小节中，描述了每个类别中的一些算法和一些模拟示例。

4.1 本地化

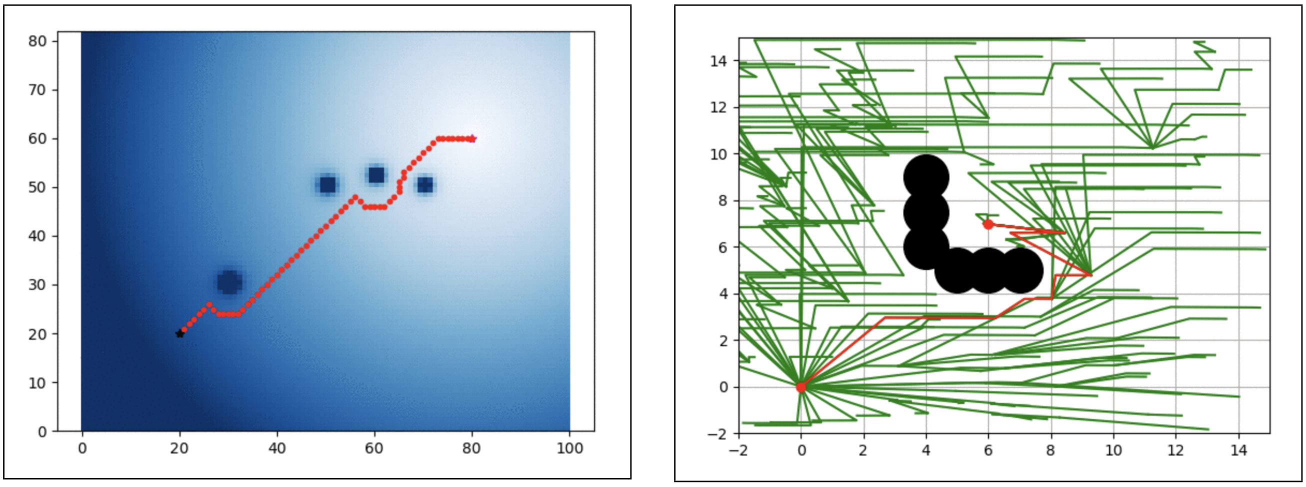
定位是机器人通过全球导航卫星系统（GNSS）等传感器了解其位置和方向的能力。在定位中，广泛使用卡尔曼滤波器、直方图滤波器和粒子滤波器等贝叶斯滤波器[31]。图 2 显示了使用直方图滤波器和粒子滤波器的定位模拟。

4.2 映射

地图绘制是机器人通过激光雷达和摄像头等外部传感器了解周围环境的能力。机器人必须识别障碍物的位置和形状才能避开它们。在地图绘制中，网格地图和机器学习算法被广泛使用[31][18]。图 3 显示了使用带有 2D 射线投射的网格映射和带有 k-means 算法的 2D 对象聚类的映射模拟结果。

4.3 SLAM

同步定位与建图（SLAM）是一种同时估计机器人姿态和环境地图的能力。 SLAM问题很难解决



Potential Field path planning

LQR-RRT* path planning[27]

Figure 5: Path planning simulation results

solve, because a map is needed for localization and localization is needed for mapping. In this way, SLAM is often said to be similar to a ‘chicken-and-egg’ problem. Popular SLAM solution methods include the extended Kalman filter, particle filter, and Fast SLAM algorithm[31]. Fig.4 shows SLAM simulation results using extended Kalman filter and results using FastSLAM2.0[31].

4.4 Path planning

Path planning is the ability of a robot to search feasible and efficient path to the goal. The path has to satisfy some constraints based on the robot’s motion model and obstacle positions, and optimize some objective functions such as time to goal and distance to obstacle. In path planning, dynamic programming based approaches and sampling based approaches are widely used[22]. Fig.5 shows simulation results of potential field path planning and LQR-RRT* path planning[27].

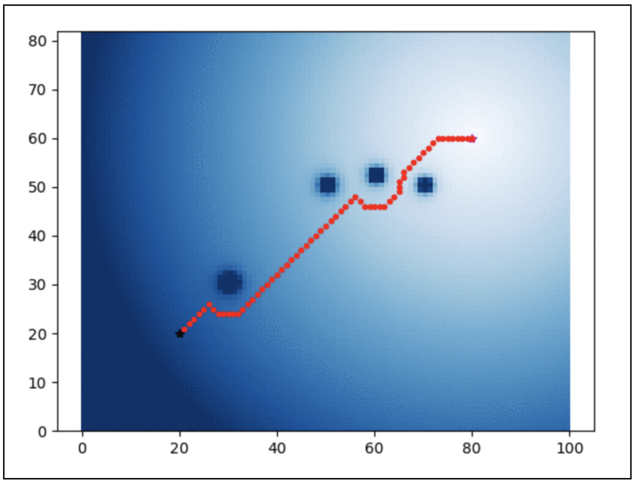
4.5 Path tracking

Path tracking is the ability of a robot to follow the reference path generated by a path planner while simultaneously stabilizing the robot. The path tracking controller may need to account for modeling error and other forms of uncertainty. In path tracking, feedback control techniques and optimization based control techniques are widely used[22]. Fig.6 shows simulations using rear wheel feedback steering control and PID speed control, and iterative linear model predictive path tracking control[27].

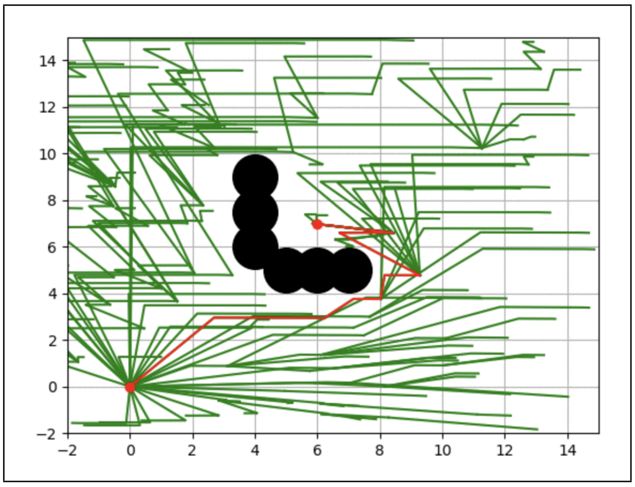
5 Conclusion and future work

In this paper, we introduced PythonRobotics, a code collection of robotics algorithms, with a focus on autonomous navigation. Related works of this project, some key ideas about this OSS project, and brief structure of this repository and simulation results were described. The future works for this project are as follows:

- Adding technical and mathematical documentation using Jupyter notebook[1].



Potential Field path planning



LQR-RRT* path planning[27]

图5：路径规划仿真结果

解决，因为定位需要地图，建图也需要定位。这样，SLAM 通常被称为类似于“先有鸡还是先有蛋”的问题。流行的SLAM求解方法包括扩展卡尔曼滤波器、粒子滤波器和Fast SLAM算法[31]。图4显示了使用扩展卡尔曼滤波器的SLAM模拟结果和使用FastSLAM2.0的结果[31]。

4.4 路径规划

路径规划是机器人搜索可行且有效的路径到达目标的能力。该路径必须满足基于机器人运动模型和障碍物位置的一些约束，并优化一些目标函数，例如到达目标的时间和到障碍物的距离。在路径规划中，广泛使用基于动态规划的方法和基于采样的方法[22]。图 5 显示了势场路径规划和 LQR-RRT* 路径规划的仿真结果[27]。

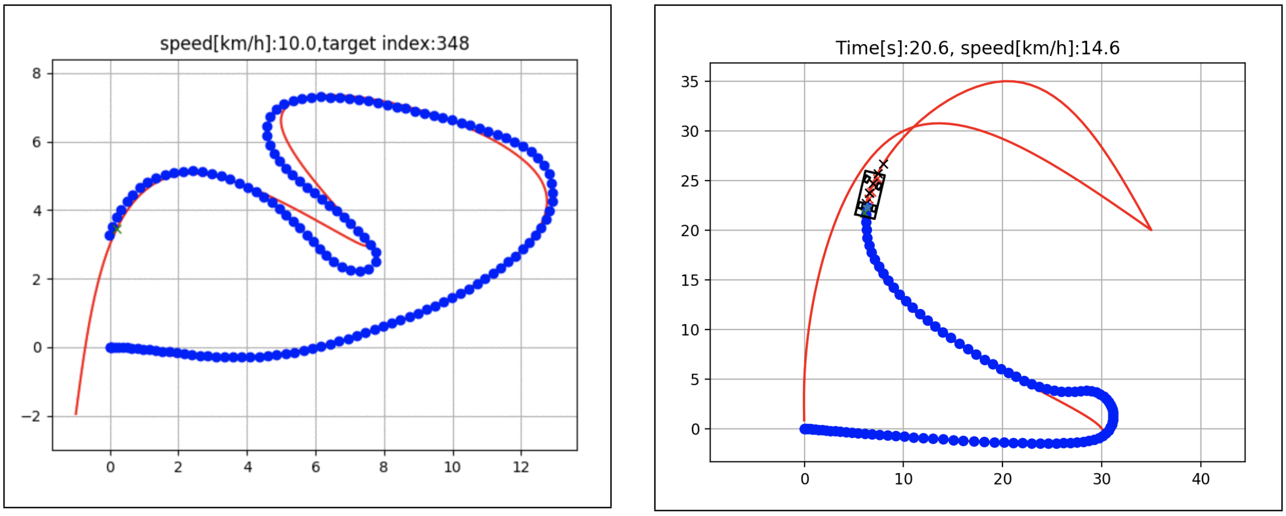
4.5 路径跟踪

路径跟踪是机器人遵循路径规划器生成的参考路径，同时稳定机器人的能力。路径跟踪控制器可能需要考虑建模误差和其他形式的不确定性。在路径跟踪中，反馈控制技术和基于优化的控制技术被广泛使用[22]。图 6 显示了使用后轮反馈转向控制和 PID 速度控制以及迭代线性模型预测路径跟踪控制的仿真[27]。

5 结论和未来工作

在本文中，我们介绍了PythonRobotics，这是一个机器人算法的代码集合，重点关注自主导航。描述了该项目的相关工作、该 OSS 项目的一些关键思想、该存储库的简要结构和模拟结果。该项目未来的工作如下：

- 使用 Jupyter Notebook 添加技术和数学文档[1]。



Rear wheel feedback steering control and PID speed control [25]

Iterative linear model predictive control

Figure 6: Path tracking simulation results

- Adding image processing samples for autonomous navigation using OpenCV[9].
- Adding simple multi-robot simulations for autonomous navigation.
- Adding a comparison to find which algorithm fits best for a specific application.

If readers are interested in these future projects, contributions are welcome. Readers can also support this project financially via Patreon[11].

6 Acknowledgments

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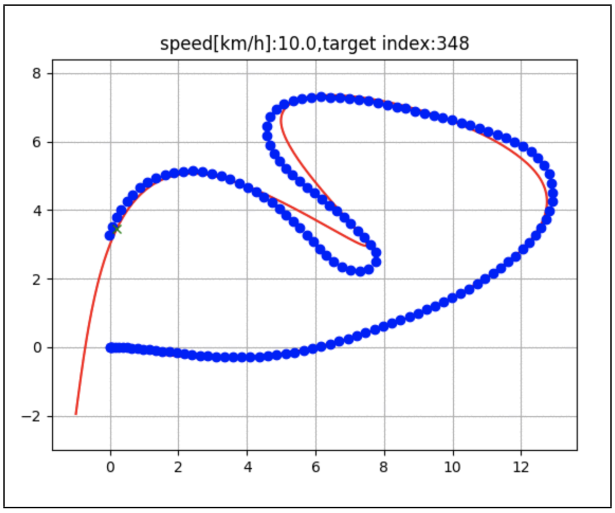
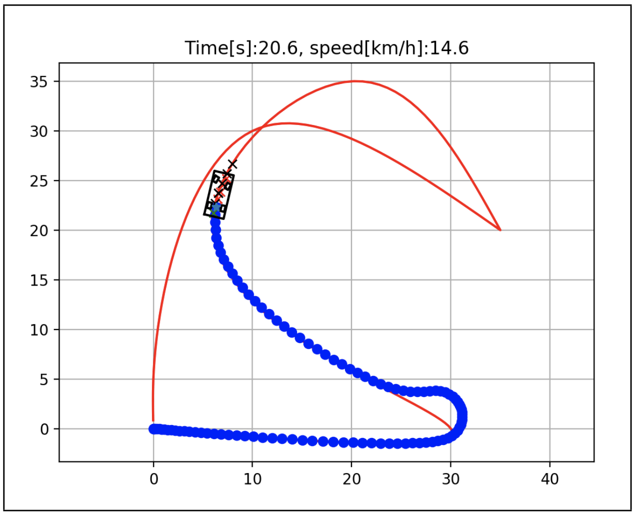
[7] Mit license. https://opensource.org/licenses/MIT.



Rear wheel feedback steering control and PID speed control [25]



Iterative linear model predictive control

Figure 6: Path tracking simulation results

- 插件 g 使用 OpenCV 进行自主导航的图像处理示例[9] 。
- 添加简单的多机器人模拟以进行自主导航。
- 添加比较以找出最适合特定应用的算法。

如果读者对这些未来的项目感兴趣，欢迎贡献。读者还可以通过 Patreon[11] 为该项目提供财务支持。

6 致谢

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